

Md Shahanur Islam Shagor

Full-Stack Developer & AI Engineer

Detail-oriented and innovative **Software Engineer** with a specialized focus on **Autonomous Systems, UAV coordination**, and **Cybersecurity**. Proven experience in developing secure, scalable web applications and research-driven projects involving **Blockchain frameworks and AI-based computer vision**. Adept at implementing **post-quantum cryptography and decentralized protocols** to solve complex technical challenges. A strong problem-solver committed to continuous learning and advancing the field of autonomous technology.

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EDUCATION

Bachelor of Science

Voronezh State University of Forestry and Technology, Russia
Computer Science and Microelectronics Engineering
2023-Continue

Diploma in Engineering

Dhaka Polytechnic Institute, Bangladesh.
Computer Technology
2016-2020

CERTIFICATE

Python Web Programming
Global IT — 2020

Java Programming
LinkedIn — 2020

Database Developer
LinkedIn — 2020

Python Programming Language
Python Institute — 2020

Python Developer
Python Institute — 2021

Java EE 7 Developer
LinkedIn — 2021

C++ Programming
CodeSignal — 2025

WORK EXPERIENCE

Web Developer (Remote)
Explo-IT | Bangladesh — 2023 - 2025

- Designed and developed scalable web applications using Laravel (backend) and Next.js (frontend).
- Built and integrated RESTful APIs for seamless data exchange between frontend and backend systems.
- Implemented authentication, role-based access control, and secure payment gateways in Laravel.
- Optimised application performance via Eloquent query tuning, caching strategies, and database indexing.
- Created responsive, SEO-friendly UIs with Next.js, Tailwind CSS, and modern JavaScript frameworks.
- Delivered real-time features (chat, notifications, live updates) using WebSockets / Pusher.
- Deployed and maintained applications on Linux servers (Nginx, Apache, Docker) with CI/CD pipelines.
- Integrated third-party services: payment gateways, social logins, and cloud storage.

Web Developer (Onsite)
DataSoft LTD | Bangladesh — 2021 - 2022

- Assisted in developing web applications using Laravel and Next.js.
- Built and maintained APIs, database queries, and responsive UI components.
- Collaborated with senior developers on bug fixing, code reviews, and feature enhancements.
- Implemented frontend layouts with Tailwind CSS and supported database migrations in Laravel.
- Wrote clear documentation and contributed to team knowledge-sharing sessions.

PROJECTS

1. WVAB: Wireless Vision-Aid System for the Blind

- Technologies: Python, YOLOv8n, ROS, NVIDIA Jetson Nano, ESP32-CAM, UDP Streaming.
- Developed an assistive navigation system utilizing YOLOv8n for real-time object detection with low-latency multilingual audio feedback.
- Optimized UDP video streaming from ESP32-CAM to Jetson Nano, achieving high frame rates for obstacle recognition in a Linux/ROS environment.

Implemented a custom spatial awareness algorithm to guide users through complex indoor environments.

SKILLS

Backend

- PHP
- Laravel
- Node.js
- Python
- Flask
- Rest APIs
- GraphQL
- WebSockets
- GO
- C++
- C#

Frontend

- React,js
- Next.js
- Vue.js
- Angular.js
- React Native
- TypeScript
- JavaScript
- Tailwind CSS
- Bootstrap
- HTML & CSS

Database

- SQLite
- Mysql
- PostgreSQL
- MongoDB
- Redis

DevOps & Tools

- Linux
- Nginx
- Apache
- Docker
- CI/CD Pipelines
- GitHub
- CMake

Mobile & Embedded

- Flutter
- Android Studio
- Raspberry Pi
- NVIDIA Jetson Nano
- ESP32-CAM
- Linux/ROS
- LiDAR

AI & Security

- YOLOv8n
- Blockchain
- Zero-Knowledge Proofs
- OWASP
- JWT / OAuth 2.0
- Post-Quantum Cryptography
- Cyber Security

Cloud & Infrastructure

- AWS (EC2, S3, Lambda)
- Google Cloud

2. Privacy-Preserving Blockchain for Autonomous Vehicles
   - Technologies: Go/Python, Zero-Knowledge Proofs (ZKP), SHA3-256, Post-Quantum Cryptography.
   - Designed a decentralized security framework for V2X (Vehicle-to-Everything) communication, focusing on identity privacy and data integrity.
   - Integrated Pedersen Commitments (ZKP) to validate vehicle transactions without revealing sensitive location or identity data.

   Researched and implemented Post-Quantum Cryptographic primitives to ensure long-term resilience against quantum-level threats.
3. SyncChat – Real-Time Communication Engine
   - Technologies: Node.js, React Native, WebSocket, WebRTC, Redis (Pub/Sub).
   - Built a high-concurrency chat engine supporting real-time messaging, live user presence, and peer-to-peer video calling via WebRTC.
   - Leveraged Redis Pub/Sub to handle message broadcasting and state management across multiple server instances.

   Developed a cross-platform mobile interface using React Native, ensuring sub-100ms message delivery latency.
4. Multivendor E-commerce Management Platform
   - Technologies: Laravel, Next.js, GraphQL, MySQL, Stripe, Nginx.
   - Engineered a scalable marketplace architecture supporting multiple vendor tiers, dynamic inventory management, and role-based access control (RBAC).
   - Integrated GraphQL for optimized data fetching, reducing frontend load times by 40%.

   Automated deployment using CI/CD pipelines and configured Nginx for load balancing and SSL termination.
5. Server Monitor — Real-Time Health Dashboard
   - Technologies: React, TypeScript, Node.js, Express, Socket.IO, Python Flask, psutil, SQLite, Docker, Nginx.
   - Developed a TypeScript + Socket.IO backend with a Python Flask agent for live CPU, RAM, and disk monitoring using psutil.
   - Implemented role-based authentication, SQLite persistence, Docker deployment, and automated health-check scheduling.

   Designed a responsive dashboard with live metrics, alert triggers, heartbeat monitoring, and failover status visualization.
6. Decentralized Coordination in Secure Autonomous Drone Swarms
   - Technologies: Python, C++, Byzantine Resilience, SHA3-256, Mesh Networking.
   - Developed a decentralized leader election and coordination framework for drone swarms, ensuring operational continuity despite node failures.
   - Implemented Byzantine-resilient algorithms and secure communication protocols using SHA3-256 chaining to prevent malicious takeovers within the swarm.

   Optimized high-concurrency swarm management using Go, focusing on secure data exchange over decentralized mesh networks.
7. Acoustic Localization for GPS-Denied Environments
   - Technologies: Python, C++, Go, CMake, Byzantine Resilience, SHA3-256, AES-256, Mesh Networking.
   - Engineered a robust localization system for autonomous UAVs operating in GPS-denied environments using acoustic signal processing.
   - Implemented TDOA (Time Difference of Arrival) algorithms to calculate precise coordinates based on sound wave propagation.

   Utilized Go for analyzing correlation matrices and noise filtering to improve localization accuracy in high-interference outdoor settings.

- Digital Ocean
- Kubernetes
- Prometheus
- Grafana

Testing & QA

- Jest
- PHPUnit
- PyTest
- Postman (API Testing)
- Supertest (Integration Testing)
- K6 / Artillery (Load Testing)

RESEARCH FOCUS

- Autonomous Drone Swarms
- GPS-Denied Navigation
- Sensor Fusion (EKF/VIO)
- Post-Quantum Cryptography
- Byzantine Fault Tolerance
- Decentralized Mesh Networks

AFFILIATIONS

- IEEE Student Member
- VSUFT Scientific Society
- Robotics & Automation Community (SMC)

HARDWARE & TOOLS

- NVIDIA Jetson Ecosystem
- Decawave UWB Systems
- RPLIDAR Infrastructure
- ROS / ROS2
- Gazebo / Pixhawk (SITL)
- CMake & GCC Toolchains
- Linux Kernel Tuning

LANGUAGES

- English (Professional)
- Bengali (Native)
- Russian (Elementary)

DEFENSE TECH INTERESTS

- Electronic Warfare (EW) Resilience
- Counter-UAS (C-UAS) Frameworks
- Anti-Spoofing & Anti-Jamming
- Tactical Mesh Formation

INTERESTS

- UAV & FPV Piloting
- Vibe Coding
- Continuous Learning
- Open-Source Contributing
- Acoustic Signal Experimentation
- Custom PC Building
- Tech Mentoring
- Technical Writing
- Hardware Benchmarking

RESEARCH & PUBLICATIONS

1. Wireless Vision-Aid for the Blind — Under Review

   Real-Time Object Detection, Multilingual Audio Feedback, and Low-Latency UDP Streaming

   - Core Contribution: Architected a $15 USD embedded assistive system integrating YOLOv8n with an offline JSON multilingual audio engine. Developed a custom chunked UDP streaming protocol for ESP32-CAM and a C++ risk-weighted navigation planner using spatial lane-scoring.
   - Key Metrics: Achieved <50{ ms} GPU latency (95.4{ ms} wireless), 15-30 FPS throughput, and 91.5% indoor navigation accuracy, delivering a 200x cost reduction over commercial solutions.

2. OmniGuard V2X Architecture — Under Review

   A Privacy-Preserving Blockchain Framework for Smart Vehicle Security.

   - Core Contribution: Engineered a 13-layer V2X blockchain framework featuring a Super Encrypted Security (SES) layer that unifies ML-KEM-512 PQC with homomorphic Pedersen commitments. Built a decentralized Go gossip swarm with mTLS REST APIs, Lamport DIDs, and MAD-Krum filtered Federated Learning.
   - Key Metrics: Hardware validation on Pi 4 + Arduino showed 5.34 ms warm-start E2E latency (under ETSI 100ms budget), 100% attack detection, and 91.7% FL accuracy under Byzantine poisoning with O(n) communication scale.

3. UAV-GDNS — Under Review

   A Heterogeneous Sensor Fusion Framework with Cooperative Swarm Intelligence.

   - Core Contribution: Architected a formally specified trilingual navigation framework decomposing real-time computation (C++20), fleet orchestration (Go), and visualization (Python). Developed an error-state Extended Kalman Filter (EKF) fusing Visual-Inertial Odometry (VIO), 12-anchor UWB TDOA multilateration, and LiDAR obstacle mapping. Implemented a V2X mesh network for distributed leader election and collision prediction.
   - Key Metrics: Achieved position error bounded at 0.18 m RMS under complete GPS denial. Sustained localization confidence above 0.78 and LiDAR obstacle clearance above 0.80 m. Swarm leader election converged within 3 rounds, delivering an average collision avoidance response time of 118 ms.

4. Multi-Layered Navigation Framework — Under Review

   A Multi-Layered Framework for GPS-Denied Autonomous Navigation in Dynamic Drone Swarm Systems.

   - Core Contribution: Engineered a heterogeneous, multi-layered framework utilizing C++20 for real-time sensor fusion, Go for fleet orchestration, and Python for telemetry visualization. Integrated tightly coupled VIO with an error-state EKF and UWB TDOA multilateration for absolute drift correction under a secure mTLS stack.
   - Key Metrics: Achieved sub-decimeter positioning with a mean localization error of 0.087 m under complete signal denial. Demonstrated successful swarm coordination of up to 16 autonomous agents executing dynamic obstacle avoidance. Successfully recovered from complete state loss within 2.3 seconds via visual loop closure and global pose-graph optimization.